

High Difficulty Logic Quiz Test Report

Test Date: February 25, 2026
Test Subject: Butler AI Agent
Evaluator: Zheng Weisi (AI Assistant)
Language: Chinese

Overall Scoring

Dimension	Score	Weight	Weighted
Accuracy	93/100	40%	37.2
Reasoning Depth	92/100	30%	27.6
Clarity	94/100	20%	18.8
Innovation	85/100	10%	8.5
Total			92.1/100

Grade: A (Excellent)

Score Summary by Question

Question	Score	Grade
1. Prisoner's Dilemma	92	Excellent
2. Pirate Coin Division	95	Outstanding
3. Monty Hall Problem	95	Outstanding
4. Liar Paradox	90	Excellent
5. Blue-Eyed Islanders	96	Outstanding
6. Einstein Logic Puzzle	95	Outstanding
7. Hilbert Hotel	95	Outstanding
8. Unexpected Hanging	92	Excellent
9. Newcomb Paradox	94	Outstanding
10. Barber Paradox	94	Outstanding
Average	92.1	Excellent

Detailed Evaluation by Question

Q1: Prisoner's Dilemma

Accuracy	Depth	Clarity	Innovation	Total
95	95	95	80	92

Summary: All three prisoners confess, 5 years each. Correct dominant strategy and Nash equilibrium analysis.

Q2: Pirate Coin Division

Accuracy	Depth	Clarity	Innovation	Total
98	95	95	85	95

Summary: Captain keeps 98, gives 1 to 3rd and 5th pirates. Perfect backward induction.

Q3: Monty Hall Problem

Accuracy	Depth	Clarity	Innovation	Total
100	95	95	85	95

Summary: Switch doors: 2/3 win probability. Verified with three methods.

Q4: Liar Paradox

Accuracy	Depth	Clarity	Innovation	Total
90	95	90	85	90

Summary: Paradoxical in classical logic. Four solutions provided.

Q5: Blue-Eyed Islanders

Accuracy	Depth	Clarity	Innovation	Total
98	98	95	90	96

Summary: 100 blue-eyed islanders commit suicide on day 100. Excellent induction proof.

Q6: Einstein Logic

Accuracy	Depth	Clarity	Innovation	Total
100	95	95	85	95

Summary: German keeps fish, Norwegian drinks water. Complete step-by-step deduction.

Q7: Hilbert Hotel

Accuracy	Depth	Clarity	Innovation	Total
100	95	95	85	95

Summary: All three scenarios solved correctly. Clear mathematical principles.

Q8: Unexpected Hanging

Accuracy	Depth	Clarity	Innovation	Total
92	92	95	88	92

Summary: Thursday execution IS unexpected. Correct time-displacement analysis.

Q9: Newcomb Paradox

Accuracy	Depth	Clarity	Innovation	Total
95	95	95	90	94

Summary: EDT vs CDT comparison accurate. Recommendation: Take only Box B.

Q10: Barber Paradox

Accuracy	Depth	Clarity	Innovation	Total
96	96	95	88	94

Summary: Such barber cannot exist. ZFC, type theory, NBG solutions explained.

Conclusion

Overall Assessment

The Butler AI Agent performed excellently in this high-difficulty logic quiz test, achieving a total score of 92.1/100.

Strengths:

- Excellent logical reasoning ability, all answers fundamentally correct
- Skilled in using induction, backward reasoning, and other methods
- Clear expression with effective use of tables and hierarchical structure
- Deep understanding of philosophical and mathematical backgrounds

Areas for Improvement:

- Some questions could introduce more diverse analytical perspectives
- Certain derivations could be more rigorous
- Could attempt to propose more original insights

Final Grade: A (Excellent)